

Partho Kumar Paul

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- Chemistry Ph.D. with six years of undergraduate teaching, including a quarter as head TA for an organic chemistry course of about 300 students (full lectures, TA management).
- Uses cooperative learning and builds free 3D tools for students (VSEPR Builder, Newman Navigator).
- Prepared to teach general, introductory, and organic chemistry, plus upper-division organic and organometallic courses.

Education

Ph.D. – Chemistry

2026

University of California, Irvine (Irvine, CA)

Defended September 10, 2026; degree expected December 2026. Advisor: Professor Jenny Y. Yang.

Dissertation: *Synthesis and Evaluation of Bio-Inspired Organic Molecules for Electrochemical CO₂ Capture and Cobalt-Hydride Mediated Transformation of Alkenes*

B.S. – Chemical Engineering

2020

University of California, Riverside (Riverside, CA)

Nanotechnology emphasis; minor in Chemistry; graduated with honors

Teaching Experience

Teaching Assistant – Department of Chemistry, University of California, Irvine

August 2021–December 2025

- Ran Organic Chemistry Laboratory sections (CHEM 51LA, 51LB) of about 20 students; gave pre-lab safety talks, supervised bench work and waste handling, taught instrument use, and graded reports and notebooks.
- Led Organic Chemistry discussions (CHEM 51A–C) of up to 50 students using cooperative learning; students worked in groups, explained their reasoning at the board, and found their own mistakes through guided questions.
- Head TA for CHEM 51A (about 300 students), Fall 2025: gave full lectures for Professor Steven Nguyen, managed the other TAs in learning the material and in grading, and handled announcements, course materials, and answer keys.
- Wrote quizzes and rubrics; graded exams for the full course (100+ students) each quarter with Gradescope.
- Evaluations: 96 of 120 respondents agreed the material was presented in a way that helped them understand it; campus TA ratings (out of 9) rose from 6.8 to 8.0 overall and from 7.3 to 8.2 for “open and fair learning environment.”

Faculty Intern – BLOOM Program

January–May 2025

Irvine Valley College (community college, Irvine, CA)

South Orange County Community College District program supporting diverse and inclusive future faculty

- Observed and analyzed Professor Steven Nguyen’s flipped-classroom approach in Organic Chemistry I.
- Created and presented two lectures on substitution reactions and one on CO₂ capture as a real-world application.
- Developed course materials and assisted with equity-focused curriculum planning.
- Developed a summer research project for community college undergraduates with Professor Nguyen; guided the students in synthesizing redox-active alloxazines and testing them by cyclic voltammetry.

Supplemental Instructor (General and Organic Chemistry) and Tutor – UC Riverside

July 2017–July 2020

- Led supplemental instruction sessions (2018–2020) with peer learning and collaborative problem-solving, and tutored students one-on-one (2017–2018); created lesson plans, evaluation criteria, and online materials.

Private Chemistry Tutor – General, Organic, and Advanced Organic Chemistry

2017–present

- Created original practice problems and worked examples; adapted explanations and pacing to each student.

Teaching Tools and Skills

[VSEPR Builder](#)

- Built a free 3D browser tool for general chemistry; students enter valence electrons, steric number, and lone pairs, each checked, before the shape appears (AX₂ through AX₂E₄).

[Newman Navigator](#)

- Built a free 3D browser tool for organic chemistry with guided questions; students pick or draw a molecule, sight down a bond, and turn the back carbon as the Newman projection, dihedral angle, and torsional energy update.

Guided Questions and Practice Materials

- Wrote guided questions that ask students to explain their reasoning and find the source of a misconception.
- Designed low-stakes practice problems, visual aids, and an organic chemistry instructional video with worked examples.

Course technology: Canvas, Gradescope, PollEverywhere, and Zoom (online sections and office hours).

Mentoring, Service, and Outreach

Research Mentoring

- Mentored three Yang Lab undergraduates in experimental design, lab safety, data analysis, and scientific writing for reports and posters; they went on to industry, medical school, and graduate school.
- Guided one mentee in writing a proposal that won a UC Irvine undergraduate research award.

Advisor – Project RISHI (student club), University of California, Irvine

Elected Representative – School of Physical Sciences

September 2024–June 2026

Associated Graduate Students (AGS) – University of California, Irvine

- Represented graduate students in university governance; debated policy proposals and contributed to discussions on TA training, student success, and equitable academic policies.
- Advocated for affordable housing, equitable stipends, mental health resources, and access to campus services; brought graduate students' feedback to university leadership.
- Worked with faculty, department chairs, and administrators to improve teaching and graduate student support.

Council on Teaching, Learning, and Student Experience

September 2024–June 2026

University of California, Irvine

- Brought graduate student perspectives to council discussions and policy debates on teaching and learning.
- Represented the needs of teaching assistants by advocating for expanded training opportunities, enhanced instructional resources, and inclusive teaching practices.

Science Outreach

- Designed and gave K–12 chemistry demonstrations to spark curiosity and broaden access to science education.

Research Experience

Graduate Student Researcher – Yang Lab, University of California, Irvine

April 2024–present

- Designed and synthesized 11 redox-active quinone and flavin targets (23 compounds with intermediates) for electrochemical CO₂ capture; built an electrochemical pH-swing CO₂ capture setup.

DOE SCGSR Fellow – Chemistry Division, Brookhaven National Laboratory (Upton, NY)

January–June 2026

- Studied electron-transfer kinetics in organic molecules by pulse radiolysis and transient absorption spectroscopy; resolved flavin semiquinone spectra and pK_a values.

Visiting Researcher – RIKEN Center for Computational Science (Kobe, Japan)

June–August 2026

- Ran TD-DFT calculations on flavin derivatives.

Graduate Student Researcher – Pronin Lab, University of California, Irvine

August 2021–March 2024

- Studied cobalt-catalyzed radical–polar crossover reactions; performed air-sensitive multistep synthesis.

Laboratory skills: recrystallization, flash chromatography, Grignard reactions, Schlenk and glovebox techniques; ¹H, ¹³C, and ¹⁹F NMR, GC-MS, and UV-Vis titrations; cyclic voltammetry and pH measurement.

Publications and Presentations

Paul, P. K.; Lin, C. C.; Yang, J. Y. Design, Synthesis, and Studies of Hydroxymethylated Quinone Derivatives for Electrochemical CO₂ Capture. Submitted.

Paul, P. K.; Yang, J. Y. Synthesis and Evaluation of Charged Flavin Derivatives for pH-Swing Mediated CO₂ Capture. In preparation.

Synthesis and Electrochemical Evaluation of Hydroxymethylated Quinones for Electrochemical Carbon Capture. ECS Meeting, May 2026.

Bridging Experiments and Computations for Mapping Electrochemical Pathways of Flavin Derivatives for Energy-Related Applications. DOE–RIKEN Workshop, Kobe, Japan, July 2026.

Honors and Fellowships

DOE Office of Science Graduate Student Research (SCGSR) Award, Brookhaven National Laboratory (2026)

DOE SCGSR International Research Collaboration Award, RIKEN, Japan (2026)

School of Physical Sciences Dissertation Year Fellowship, UC Irvine (Summer 2025)

Eugene Cota-Robles Fellowship, UC Irvine (first and fourth year)

Bourns Foundation Engineering Scholarship, UC Riverside (2019)

Poster Award, UC Irvine Undergraduate Research Symposium in Chemistry (2019)

UC LEADS Scholar, UC Riverside (2018)